

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Microsoft SAT 80, TX -- station KHD0, America/Chicago
Committed pair OSM 1533392813 -> 1533392812
 Microsoft SAT 81 / Microsoft SAT 80
Facade gap 74.4 m between the two halls
Weather record 43,148 real hourly records from KHD0
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise
 0.4822 C at 125 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2022-02-13 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. The reactive incumbent took 14 hours -- 11 more -- but declared 0 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 15 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 14 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
 the incumbent broke its own switch budget 1 time(s) to stay safe
 15 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	-1.819	18.0	-2.220	1.600
01	mechanical	yes	switch budget	-1.290	18.0	-1.110	1.434
02	mechanical	yes	switch budget	-1.675	18.0	-1.718	1.214
03	mechanical	yes	switch budget	-2.077	18.0	-3.280	1.444
04	mechanical	yes	switch budget	-1.192	18.0	-2.984	1.314
05	mechanical	yes	switch budget	-1.497	18.0	-2.984	1.179
06	mechanical	yes	switch budget	-3.838	18.0	-5.610	0.969
07	mechanical	yes	switch budget	-1.722	18.0	-2.984	1.233
08	mechanical	yes	switch budget	4.189	18.0	4.441	2.269
09	mechanical	yes	switch budget	7.763	18.0	9.450	3.113

10	mechanical	yes	switch budget	10.873	18.0	12.128	3.331
11	mechanical	yes	switch budget	14.139	18.0	13.804	2.209
12	mechanical	yes	switch budget	16.553	18.0	15.296	1.712
13	mechanical	yes	switch budget	17.675	18.0	16.584	1.429
14	mechanical	no	dry-bulb	18.776	18.0	17.691	1.518
15	mechanical	no	dry-bulb	19.135	18.0	17.694	1.476
16	mechanical	no	dry-bulb	25.404	18.0	31.312	1.234
17	mechanical	no	dry-bulb	25.020	18.0	29.914	1.519
18	mechanical	yes	-	15.318	18.0	11.509	1.844
19	mechanical	no	dry-bulb	19.960	18.0	8.729	1.801
20	mechanical	no	dry-bulb	18.127	18.0	8.181	1.831
21	FREE	yes	-	7.943	18.0	4.736	2.068
22	FREE	yes	-	5.718	18.0	2.516	1.960
23	FREE	yes	-	3.911	18.0	0.000	1.447

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.819 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.290 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.675 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.077 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.192 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.497 C

against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.838 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.722 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.189 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.763 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.873 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.139 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.553 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

13:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.675 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1

changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.776 C against a 18.0 C limit, so it fails by 0.776 C. It would take a limit 0.776 C higher, or a bound 0.776 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.135 C against a 18.0 C limit, so it fails by 1.135 C. It would take a limit 1.135 C higher, or a bound 1.135 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.404 C against a 18.0 C limit, so it fails by 7.404 C. It would take a limit 7.404 C higher, or a bound 7.404 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.020 C against a 18.0 C limit, so it fails by 7.020 C. It would take a limit 7.020 C higher, or a bound 7.020 C tighter, to change this hour.

18:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.318 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.960 C against a 18.0 C limit, so it fails by 1.960 C. It would take a limit 1.960 C higher, or a bound 1.960 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.127 C against a 18.0 C limit, so it fails by 0.127 C. It would take a limit 0.127 C higher, or a bound 0.127 C tighter, to change this hour.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.943 C, which is 10.057 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.068 C, and the margin is measured, not chosen: 1.641 C of group-conditional forecast error for this hour of day, plus 0.4266 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.718 C, which is 12.282 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.960 C, and the margin is measured, not chosen: 1.534 C of group-conditional forecast error for this hour of day, plus 0.4266 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 3.911 C, which is 14.089 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.447 C, and the margin is measured, not chosen: 1.151 C of group-conditional forecast error for this hour of day, plus 0.2955 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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